Source-Native Benchmarking for Nature-Inspired Time-Series Families: A Fail-Closed Experimental Protocol

LumenCore Technical Note - Current Public-Safe Review Draft

This technical note reports local software, custody, benchmark, and protocol states. It is not peer review, independent validation, a performance claim, field validation, trading alpha, realized savings, enterprise value, a patent opinion, or deployment authority.

Abstract

LumenCore registers candidate computational families inspired by natural forms, but does not treat inspiration as evidence. This note reports a source-native benchmark ledger, the retrospective disposition of prior leads, a fixed-rule 12-pair retrospective market panel, and a frozen prospective protocol. The current ledger contains 140 registered families, 35 implementations, and 126 direct candidate-source-baseline comparisons. The market panel repairs the prior single-series bookkeeping bottleneck and contains one narrow exploratory Holm-positive comparison, but its candidate loses on mean to three other registered baselines. No candidate passes the promotion gate. The scientific contribution is therefore a reproducible comparison and falsification framework, not a performance champion.

Research Question

Can a predeclared candidate family beat every accepted baseline for a specific source, series, cadence, and horizon under prospective custody, clustered inference, an effect floor, and familywise error control?

Current Evidence Snapshot

State	Count	State	Count
Registered families	**140**	Market comparisons inferentially insufficient	**48**
Implementations present	**35**	Kraken panel pairs	**12**
Missing implementations	**105**	Kraken panel comparisons	**16**
Direct candidate-source cards	**23**	Kraken panel exploratory Holm positives	**1**
Direct comparisons	**126**	Kraken panel all-baseline mean winners	**0**
Global Holm-positive comparisons	**0**	Kraken panel promotions	**0**
Promoted champions	**0**	Prospective eligible observations	**0**
Market-signal comparisons	**48**		

1. Source-Native Experimental Design

Why source-native baselines matter

A candidate can appear strong when it is compared against an inappropriate cadence, horizon, seasonal period, or baseline. This protocol therefore treats the source contract as part of the hypothesis. A family must win within each named source against the complete accepted baseline roster before any broader promotion.

Evaluation controls

- Candidate-by-source cards evaluated against the baseline roster registered for that source, cadence, series, and horizon.
- Expanding history ends immediately before each forecast origin.
- Inference clusters overlapping origins and horizons at the source-series level instead of treating every row as independent.
- Global Holm correction across the candidate-source-baseline comparison family; prior subset leads receive no preference.
- Positive, neutral, negative, inconclusive, and invalid outcomes are retained.

Registered baseline roster

- naive_last	- drift
- moving_average	- exponential_smoothing
- linear_trend	- seasonal_naive_source_period
- damped_holt_ets	- autoregressive_ridge_source_lag

Retrospective result

The former FRED and TWELVE_DATA subset leads are retired because they do not survive source-series clustering and the complete source-native baseline gauntlet. Four fixed market-signal families were also run against four registered baselines on each of Kraken, TwelveData, and AlphaVantage. All 48 market comparisons are inferentially insufficient because each source currently supplies one source-series cluster. A separate fixed-rule Kraken panel then evaluated the same four candidates against four baselines across 12 pre-scoring-selected pairs. One of 16 comparisons was positive after exploratory global Holm correction: beast_strategy_trend versus ridge_return_baseline, with mean unannualized risk-adjusted-score delta 0.061277644465, raw exact sign-test p 0.00634765625, and global Holm-adjusted p 0.04443359375. The same candidate lost on mean to buy-and-hold, moving-average-cross, and volatility-targeting baselines. Across the authoritative ledger and panel, zero candidate cards cleared the complete registered baseline set and zero candidates pass promotion.

Current conclusion: no alpha, edge, or champion is established.

2. Frozen Prospective Protocol

Candidate identity

The registry label is **fractal_brownian_surface**. The scientific estimator is **hurst_conditioned_multiscale_increment_heuristic_v1**: Hurst-conditioned weighted multiscale increment heuristic using lags 1, 2, 4, and 8 with a bounded persistence multiplier. The estimator name prevents the registry label from being mistaken for a validated fractional Brownian motion model.

Primary endpoint and multiplicity

The endpoint is relative_mean_absolute_error (rMAE = MAE_candidate / MAE_baseline). The candidate must clear 16 predeclared contrasts under one_sided_holm_familywise at familywise alpha 0.05. The maximum accepted rMAE is 0.95, corresponding to at least 5 percent relative improvement.

Future-only sample gates

- Daily arm: at least 520 future eligible sessions per series and 104 nonoverlapping five-session blocks.
- Monthly arm: at least 60 future releases per series.
- Uncertainty: synchronized_moving_block_bootstrap with 20000 frozen replications.
- Sample gates cannot be reduced after freeze.

Current protocol state

Protocol	State
LUMENCORE_TS_ SOURCE_NATIVE_20260729_V1	FROZEN_AWAITING_FUTURE_OBSERVATIONS
Promotion decision	WAITING_FOR_NEW_SOURCE_ROWS
Eligible future observations	0

Decision states

- **replicated_lead**: All three former lead contrasts must pass inside the full 16-test Holm family and each adjusted one-sided bound must clear the 5 percent effect floor.
- **source_winner**: The candidate must beat all eight source baselines, no series-horizon cell may be worse by more than 5 percent, and the p95 error ratio must not exceed 1.10.
- **cross_source_promotion**: A source winner must repeat in a second independent prospective period.
- **integrity_failure**: INVALID
- **insufficient_sample**: INCONCLUSIVE_WAITING_FOR_NEW_SOURCE_ROWS
- **primary_failure**: CORRESPONDING_CLAIM_FALSIFIED

3. Contribution, Limits, and Reproducibility

Scientific contribution

- A source-native baseline contract that prevents cross-source or cadence-mismatched promotion.
- A custody gate that binds exact source snapshots before benchmark acceptance.
- A clustered inference rule that avoids pseudoreplication from overlapping forecast origins and horizons.
- A full-family multiple-testing rule that prevents cherry-picking isolated wins.
- A future-only protocol with fixed endpoints, effect floor, sample gates, ablations, and falsification states.
- A cost-aware market-signal replay that uses identical timestamps, future-return rows, turnover costs, and source-specific baselines without granting a promotion from descriptive wins.
- A pre-scoring 12-pair Kraken panel that holds candidate, baseline, timing, and 10-basis-point turnover-cost rules fixed while retaining losses and a narrow exploratory positive result.
- A machine-readable claim boundary that keeps software proof separate from field, economic, or deployment claims.

Limitations

- 105 of 140 registered families lack implementations.
- Only 3 lanes currently have executable direct measured adapters; the wider nature-inspired registry remains inventory, synthetic stress, or context until implemented.
- The original 48 market-signal comparisons remain inferentially insufficient under the predeclared five-cluster minimum because each source has one registered series.
- The 12-pair Kraken panel meets the exploratory pair-count floor, but pair-level signs share one exchange and overlapping market timestamps. Independence is therefore unconfirmed, and its one narrow Holm-positive comparison is not confirmatory alpha or edge.
- The panel's narrow trend-versus-ridge result is not a promotion: the same candidate loses on mean to the other three registered baselines, and no candidate clears the complete four-baseline set.
- The prospective protocol has zero eligible future observations and cannot yet support a prospective accuracy conclusion.
- No result establishes universal superiority, field performance, trading alpha, realized savings, customer acceptance, or deployment authority.

Reproducibility receipts

The ledger, prospective configuration, prospective status, and retrospective market panel are bound below. Hashes establish byte identity and custody only; they do not establish scientific correctness or external acceptance.

Artifact	Bytes	SHA-256
out/ops/source_native_family_baseline_ledger_latest.json	2244585	627d0add4d31514f7a47dbf8e8c51b0fa46df72635f336133f54abb85b00927a
config/time_series_source_native_prospective_protocol_v1.json	9189	ede86cb35b5edebbeccdc318336148b9b6ab83d1ad15907e241e3086e5b00800
out/ops/time_series_source_native_prospective_protocol_status.json	1108	535f8494caa6ef409e44bdc901b0423c287ff578bbc5876a2b48ec96d5f7e37d
out/ops/market_signal_kraken_panel_benchmark_latest.json	318407	02a8b81e5b0a34073cb3140a1f8f7359de93fc8bb0ddef23152709252fbd97c2

Legacy concept-paper disposition

Three older papers are preserved as historical speculative concepts and blocked from upload. Their BioGeometry, scalar-field, bioresonance, cooling-savings, zero-point, weather-control, and wormhole-adjacent claims are not supported by the current source-native benchmark evidence.

Whitepaper payload SHA-256: df603bae68cc4d053f0956d44eda0a4997e5c2a2cf8b992cd7871e2cf424e3a2